

INDIAN STATISTICAL INSTITUTE

Project Linked Person — Selection 2026–2027

Practical Examination

Total: 14 + 16 + 10 = 40 marks Duration: 90 minutes

EXAMINATION INSTRUCTIONS

1. You have root on a virtual machine. It is connected to no real network. Log in with `ssh candidate@192.168.56.10`.
2. Each question has four parts: *the problem*, *how to reproduce it*, *what to fix*, and *how to check your work* (so that you do not have to guess whether you are finished).
3. Partial credit would be given if you can explain the problem and the reason behind, even if you cannot fix it.

Q1. Disk full on the lab file server.

[14 marks]

Reported by Debarshi, research scholar: “I cannot copy anything into `/srv/data`, it says the disk is full, and nothing I can see in there accounts for that much space. Submitting a job into `/srv/spool` also says the disk is full, but `df -h` shows that one has plenty of space left. That part makes no sense to me.”

Background: this server has three separate disks. `/srv/data` and `/srv/spool` look like ordinary folders, but each one is really its own small disk, attached at that path (a *mount point*). The rest of the system lives on a third, much larger disk.

<code>/</code>	the system disk	about 24 GB
<code>/srv/data</code>	the project data disk	1 GB
<code>/srv/spool</code>	the job-submission disk	48 MB

Though the two complaints look alike, they are **two unrelated faults on two different disks**, with different causes. Fixing one does nothing for the other.

(a) `/srv/data` is full

[11 marks]

`/srv/data` is at 100%, and the files you can see there do not add up to the space reported.

Reproduce:

<code>df -h /srv/data</code>	# 100% used
<code>sudo du -sh /srv/data</code>	# much smaller than df says is in use

Fix: bring `/srv/data` back to **under 10% used** (the `Use%` column of `df -h`). There is more than one reason this disk is full: freeing the largest files may not work. Observe the differences shown in the output of `df` and `du`. Explain the reason behind the difference.

Check:

<code>df -h /srv/data</code>	# Use% under 10%
<code>ls /srv/data/current</code>	# five survey-block CSV files, still there

(b) /srv/spool refuses new files

[3 marks]

New files cannot be created on `/srv/spool`, *although `df -h` reports free space on it*. Something other than free space has run out. This has nothing to do with `/srv/data`.

Reproduce:

```
df -h /srv/spool          # shows space free
touch /srv/spool/incoming/testjob  # "No space left on device"
```

Fix: make `/srv/spool` able to accept new files again. The current problem is not due to a lack of free space in the disk. Explain why `/srv/spool` reported “No space left on device” while `df -h` showed free space.

Check:

```
touch /srv/spool/incoming/t && echo OK && rm /srv/spool/incoming/t
```

Constraints

- Nothing under `/srv/data/current` may be deleted or changed.
- `/srv/data/archive/README.txt` says what is safe to remove there. Read it before deleting anything.

Q2. Shared project folder unusable.

[16 marks]

Reported for the ACMU Lab group: “Buddhadev cannot save anything into `shared/` which he could last month. And Chandrima from the internal audit cell needs to *read* that folder for the annual review, and cannot see anything at all.”

Background: who is who. `/srv/projects/acmu/` (and `shared/` inside it) belongs to the `acmu` group. Anyone who is neither a member nor the auditor should have no access at all.

arnab	member of <code>acmu</code>	read and write everything
buddhadev	member of <code>acmu</code>	read and write everything
chandrima	auditor, not a member	read everything, write nothing

To see the current state (need `sudo` for some of these):

```
cd /srv/projects/acmu
stat -c '%a %U:%G %n' . shared shared/notes.md
id buddhadev
getfacl /srv/projects/acmu/shared
```

Underneath the two complaints are **three independent faults**, each with its own cause and fix. **Making the fixes will need root (sudo)**, however, perform the **test as the users, not as root**: root can do everything, so a check run as root proves nothing.

(a) Buddhadev is no longer in the `acmu` group

[5 marks]

Reproduce: Check `id buddhadev` where `acmu` is missing. *Fix:* put him back, without disturbing his other groups. Keep checking with `id buddhadev` until it shows `acmu` again.

(b) The `shared` folder does not let the group write

[8 marks]

Members of `acmu` cannot create files in the folder at all. Arnab is still a member, so test as `arnab`:

```
sudo -u arnab touch /srv/projects/acmu/shared/t1      # Permission denied
stat -c '%a %U:%G %n' /srv/projects/acmu /srv/projects/acmu/shared
```

Fix: both directories must let the acmu group read, write and enter them, and give everyone else no write access.

```
sudo -u arnab touch /srv/projects/acmu/shared/t1 && echo "write OK"
sudo rm -f /srv/projects/acmu/shared/t1
```

(c) The auditor cannot read the folder [3 marks]

Fix: grant read-only access to everything under /srv/projects/acmu/, including files created there in future, and do it *without* adding chandrima to the group.

```
sudo -u chandrima cat /srv/projects/acmu/shared/notes.md # must work
sudo -u chandrima touch /srv/projects/acmu/shared/t2      # must FAIL
```

When all three are done

Buddhadev's complaint needs (a) and (b) together:

```
sudo -u buddhadev touch /srv/projects/acmu/shared/t3      # must work
sudo -u buddhadev cat /srv/projects/acmu/shared/notes.md  # must work
```

Constraints

- **Chandrima must not be put into the acmu group.** Audit policy forbids putting auditors into the groups they audit: read everything, write nothing.
- No chmod 777, anywhere. This is a multi-user machine.

Q3. Cannot log in to the ACMU lab server. [10 marks]

Reported by Debarshi, research scholar: "I am supposed to log in to the lab server as acmusrv. It worked last week. Now it just says *Permission denied*, and then asks me for a password I was never given. I have not touched anything since."

ssh acmulab should log you in to the account **acmusrv** on the lab server using the key ~/.ssh/id_acmu, **without a password**. Two things are wrong, and they produce the **same message**: Permission denied (publickey). Identify both of them.

How to reproduce it

```
ssh acmulab
cat ~/.ssh/config
ls -l ~/.ssh/
```

What to fix [10 marks]

3.1 Make the acmulab alias use the right key. The ~/.ssh/config file is corrupted.

Fix it so that ssh acmulab succeeds without a password prompt.

3.2 Make the key usable. Find out why ssh still refuses to use the key, and fix it.

How to check you have fixed it

```
ssh acmulab whoami          # prints: acmusrv
ls -l ~/.ssh/id_acmu        # -rw----- (0600)
```

If that prints acmusrv without asking you for anything, the key is doing the work.